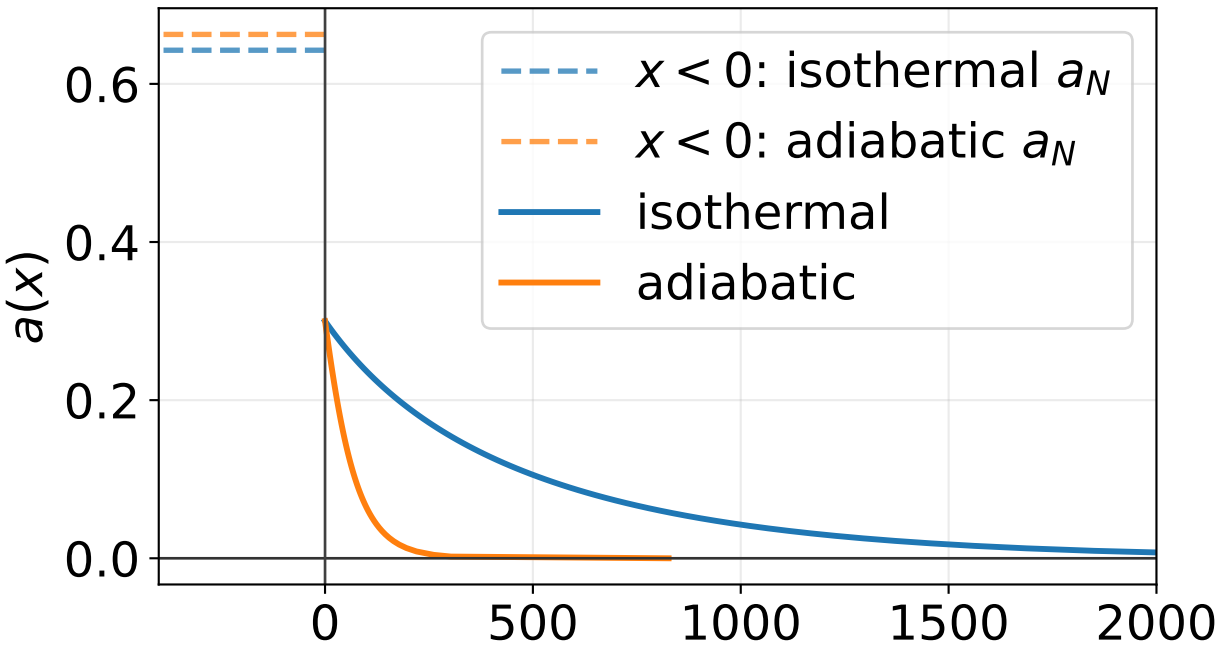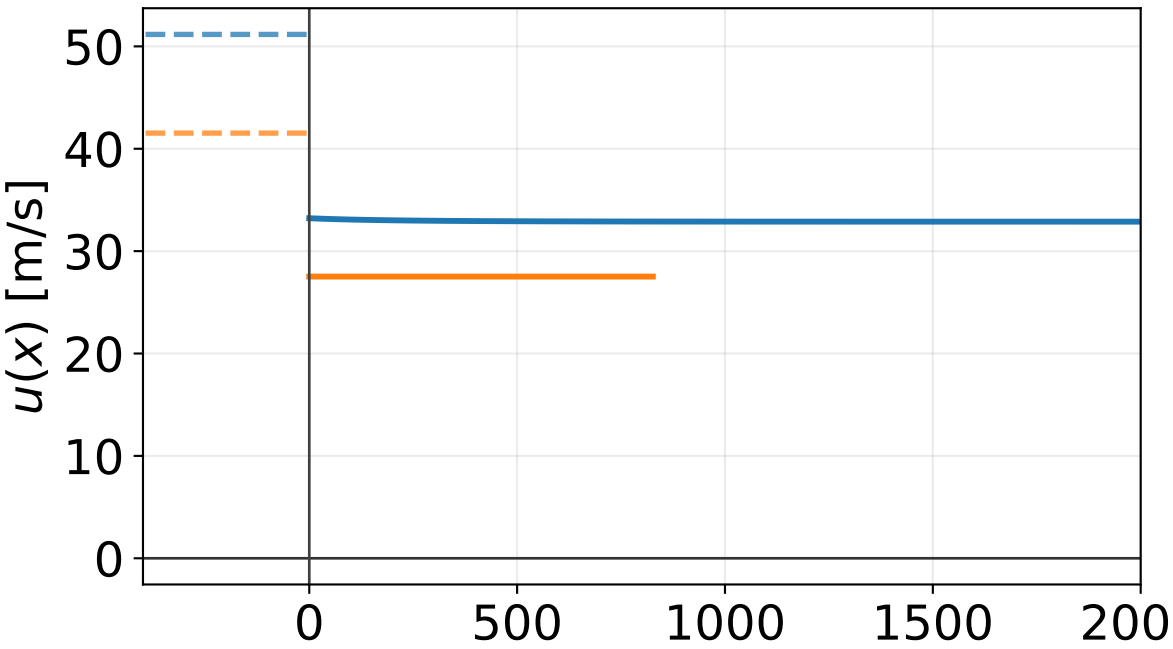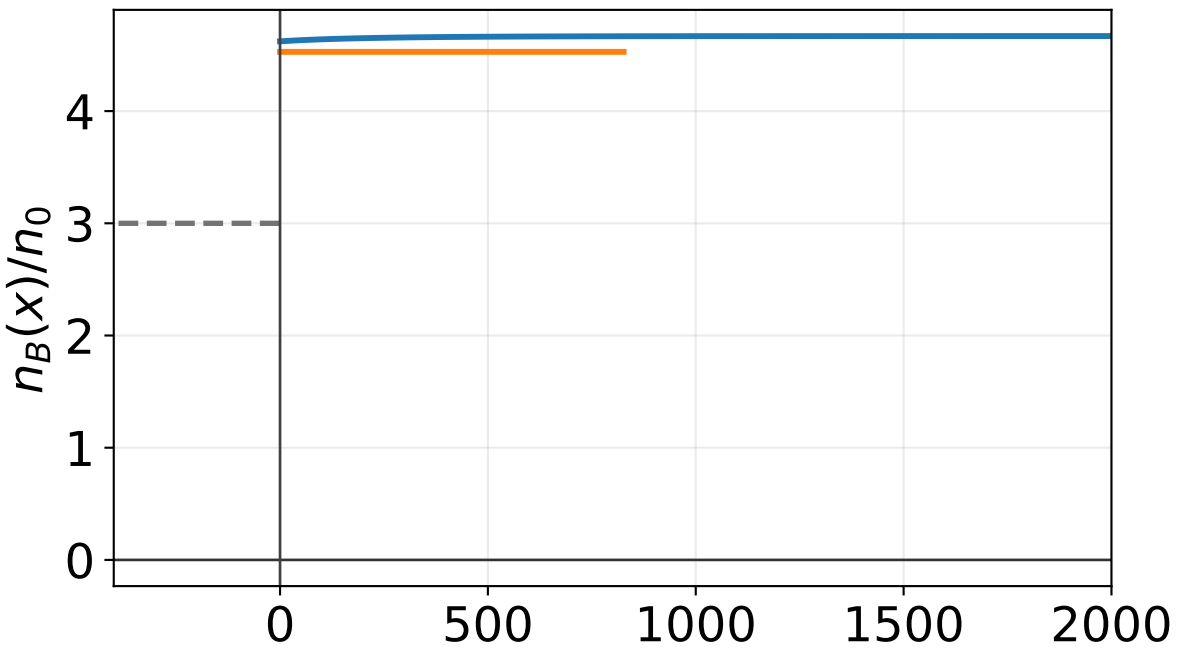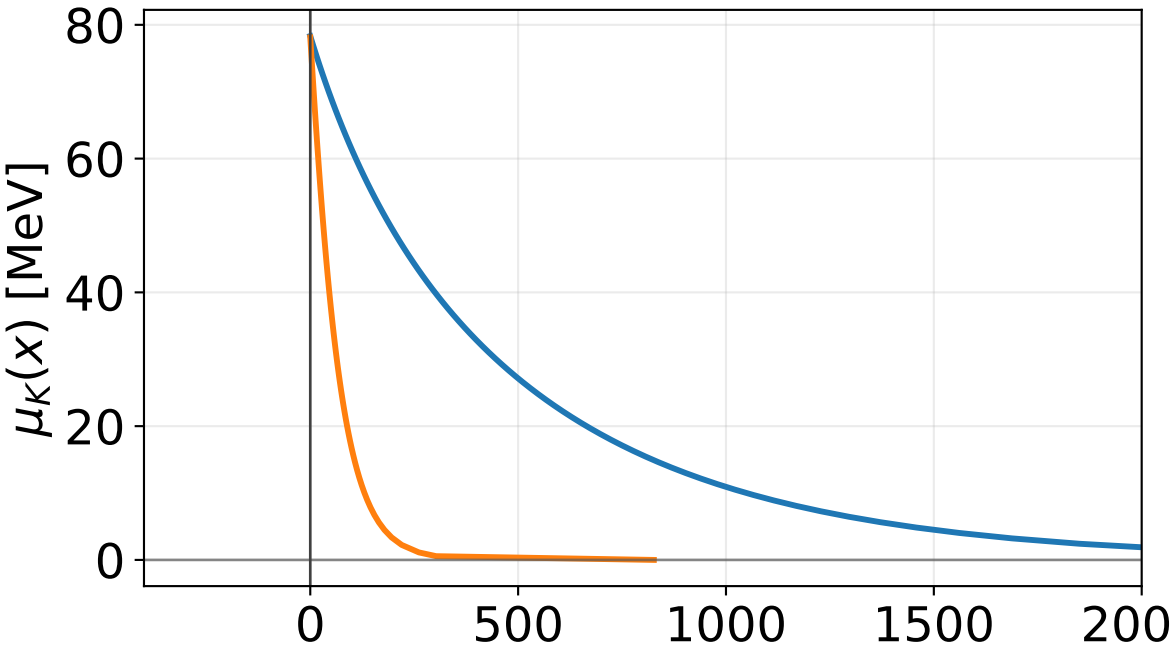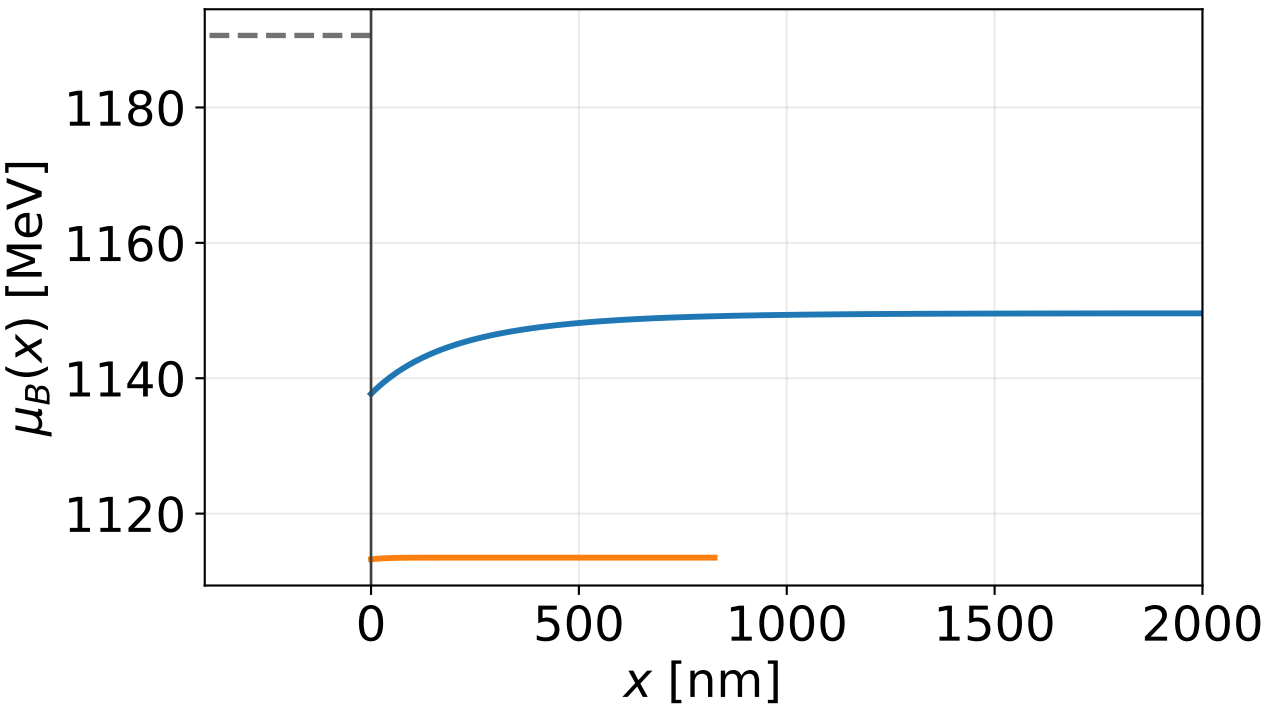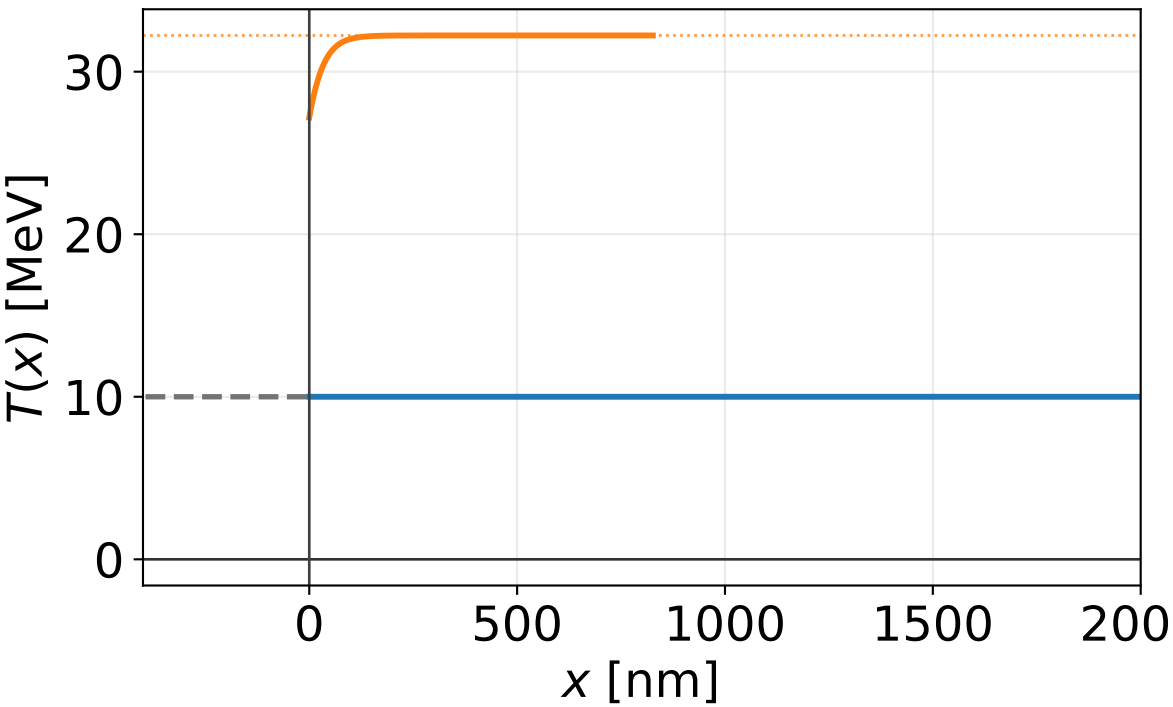Isothermal vs Adiabatic Comparison: $T = 10.00$ MeV, $n_B^N = 3.0\,n_0$, $a_{Q\star} = 0.300$